

MODEL-10

Total Hydrocarbon Analyzer

Product Data Sheet



Overview

The VIG Industries, Inc. Model-10 is a microprocessor-based, spot-heated total hydrocarbon (THC) gas analyzer designed for high accuracy, sensitivity, and stability. The analyzer uses a flame ionization detector (FID) for continuous hydrocarbon measurement in gaseous mixtures or ambient air. The detector is spot heated to prevent sample condensation effect and support repeatable, reliable analytical performance across a broad range of hydrocarbon concentrations.

Features

- Automatic start-up/ignition
- Two-stage sample filter with exchangeable sintered stainless-steel elements
- Automatic fuel shut-off system
- Automatic flame-out indicator
- Adjustable alarm settings
- Automatic ranging
- Precision 1.5% of full scale
- 19 in. rack/bench mount
- Choice of hydrogen or hydrogen/helium fuel

Options

- Automatic calibration
- Zero and calibration solenoids with software
- RS-232 interface
- Second concentration level alarm
- Internal combustion air supply
- Purge and internal cleaning system
- Dilution system
- User-selectable fuel: H₂ or H₂/He mixture
- Range recognition relays
- Multi-point sequencer
- Sample shut-off

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Standard Specifications

Measuring method	Spot heated flame ionization detector (FID)
Measurement ranges	Four ranges per analyzer. Standard ranges: 0-10 0-100 0-1000 0-10000 ppm with 0.01 ppm lower detection limit; or 0-100 0-1000 0-10000 0-100000 ppm with 0.1 ppm lower detection limit. Other ranges are available upon request.
Zero and span noise	Less than 0.5% of full scale
Zero and span drift	+/- 1.5% full scale per 24 hours
Linearity	Within 1.5% of full scale through all ranges
Repeatability	Within 1.5% of full scale through all ranges
Stability	Within 1.5% of full scale through all ranges
Oxygen synergism	Within 1.5% of full scale within selected range
Response time	Within 5 seconds to 90% of final reading
Ambient temperature	50 degrees F to 120 degrees F
Flow rate	Standard: 4 L/min standard; or 10 L/min upon request
Physical dimensions	19 in. wide front panel; 16.75 in. wide chassis; 18 in. deep chassis; 21 in. deep with fittings and handles; 7 in. high
Weight	35 - 45 lbs, depending on options
Safety	Flame-out indicator lamp; flame-out alarm contact on back panel; fuel shut-off; calibration and zero solenoid shut-off; optional sample shut-off
Voltage output	Standard: 0-10 VDC; or 0-1 VDC or 0-5 VDC upon request
Current output	4-20 mA, sourcing
Flame-out alarm	Normally open, low-current relay contact; closes on alarm; latching
Concentration 1 alarm	Normally open, low-current relay contact; closes on alarm; latching
Ignition	Automatic; can be set to manual by operator from front panel
Glow plugs	Main and spare glow plugs installed; selectable by switch on back panel

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Warm-up time	Usable in approximately 45 minutes; stable in approximately 2 hours
Display	Graphic, backlit, 240W x 64H pixels, high contrast, wide viewing angle

Operation Requirements

Fuel	UHP hydrogen at 18 psi incoming pressure standard; or UHP hydrogen/helium mixture at 18 psi incoming pressure optional
Combustion air	Oil/water/hydrocarbon-free instrument air at 18 psi incoming pressure
Zero calibration gas	UHP zero grade air or nitrogen at 9 psi incoming pressure
Span calibration gas	Known concentration of operator-selected hydrocarbons balanced in either air or nitrogen at 9 psi incoming pressure
Optional gases	UHP zero grade air or nitrogen for units with dilution option at 18 psi incoming pressure; oil/water/hydrocarbon-free instrument air for cleaning options at 50 psi incoming pressure
Power requirements	115VAC at 60 Hz at 720 W; or 220 VAC at 50 Hz upon request

Warranty

All instruments sold by VIG Industries, Inc. are warranted for a period of one (1) year from date of purchase against defects in materials and workmanship. The seller warrants that the product supplied conform to the specifications assigned thereto. There is no other warranty either expressed or implied. Seller liability is limited specifically to the cost or assigned value of the items sold. Service contracts are available after the warranty expires.